

Report: Weak value extraction in a three-path neutron interferometer

1 Path weak-values in a three-path interferometer

Let us consider a three-paths interferometer with initial state

$$|\psi(\phi_1, \phi_2, \phi_3)\rangle = \frac{e^{i\phi_1} |1\rangle + e^{i\phi_2} |2\rangle + e^{i\phi_3} |3\rangle}{\sqrt{3}} = |\psi(\vec{\phi})\rangle, \quad (1)$$

where $|j\rangle$ and ϕ_j , with $j = \{1, 2, 3\}$, represent the quantum state of path j and its initial phase, respectively. For simplicity, the initial phases are indicated with the vector $\vec{\phi} = (\phi_1, \phi_2, \phi_3) = \sum_j \phi_j \vec{\delta}_j$, with $\{\vec{\delta}_j\}$ representing the standard vector basis in $\mathbb{R}^3$. Next, an additional phase shifts χ_j can be induced in any of the paths, transforming the initial state according to $|\psi(\vec{\phi})\rangle \rightarrow e^{-i(\chi_1 \hat{\Pi}_1 + \chi_2 \hat{\Pi}_2 + \chi_3 \hat{\Pi}_3)} |\psi(\vec{\phi})\rangle$, where $\hat{\Pi}_j = |j\rangle \langle j|$ are the path projectors. Finally, at the last beam-splitter of the interferometer, the transformed state is projected onto the final state

$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle + |3\rangle}{\sqrt{3}}. \quad (2)$$

By defining the vector of induced phase shifts $\vec{\chi} = \chi_1, \chi_2, \chi_3$, we can express the intensity measured at the exit port of the interferometer as

$$\begin{aligned} I(\vec{\phi} + \vec{\chi}) &= A \left| \langle \psi_+ | e^{-i(\chi_1 \hat{\Pi}_1 + \chi_2 \hat{\Pi}_2 + \chi_3 \hat{\Pi}_3)} |\psi(\vec{\phi})\rangle \right|^2 \\ &= \frac{A}{3} \left[1 + \frac{2}{3} \cos(\phi_1 + \chi_1 - \phi_2 - \chi_2) + \frac{2}{3} \cos(\phi_1 + \chi_1 - \phi_3 - \chi_3) + \frac{2}{3} \cos(\phi_2 + \chi_2 - \phi_3 - \chi_3) \right] \end{aligned} \quad (3)$$

where A is a constant that depends on the setup characteristics and has the unit of count rate.

The path weak-values w_j , characterized by the initial state $|\psi(\vec{\phi})\rangle$, the path observables $\hat{\Pi}_j$ and the final state $|\psi_+\rangle$, are complex quantities defined as

$$w_j = \frac{\langle \psi_+ | \hat{\Pi}_j | \psi(\vec{\phi}) \rangle}{\langle \psi_+ | \psi(\vec{\phi}) \rangle} = w_j^R + i w_j^I, \quad (4)$$

where the terms w_j^R and w_j^I are, respectively, their real and imaginary parts. As shown in [1], the measured intensity for the a phase shift $\chi_j \vec{\delta}_j$ induced only in path j can be expressed as

$$\begin{aligned} I(\vec{\phi} + \chi_j \vec{\delta}_j) &= A \left| \langle \psi_+ | e^{-i\chi_j \hat{\Pi}_j} |\psi(\vec{\phi})\rangle \right|^2 \\ &= A \left| \langle \psi_+ | \psi(\vec{\phi}) \rangle \right|^2 \left[1 + 2(|w_j|^2 - w_j^R)(1 - \cos \chi_j) + 2 w_j^I \sin \chi_j \right] \end{aligned} \quad (5)$$

and the weak values can be extracted as

$$w_j^R = \frac{I_j}{I(\vec{\phi})} + \frac{I(\vec{\phi}) - I(\vec{\phi} + \pi \vec{\delta}_j)}{4I(\vec{\phi})}, \quad (6)$$

$$w_j^I = \frac{I(\vec{\phi} + \frac{\pi}{2} \vec{\delta}_j) - I(\vec{\phi} - \frac{\pi}{2} \vec{\delta}_j)}{4I(\vec{\phi})}, \quad (7)$$

with $I_j = A \left| \langle \psi_+ | \hat{\Pi}_j | \psi(\vec{\phi}) \rangle \right|^2$.

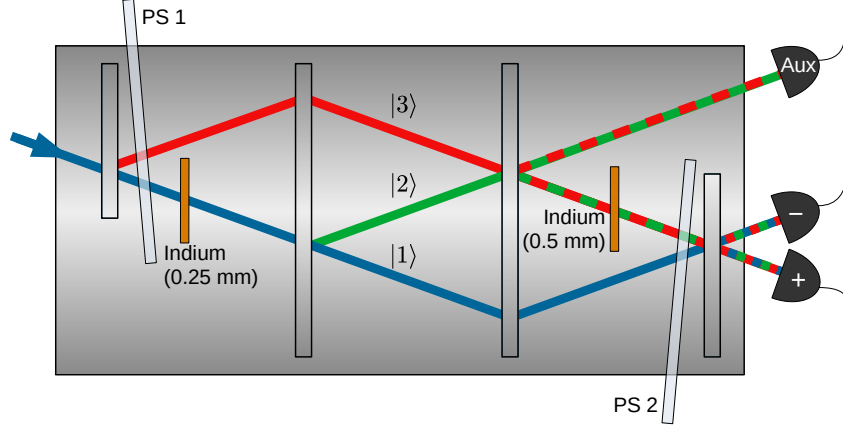


Figure 1: Scheme of the setup

2 Setup

A scheme of the setup is shown in Fig.1. Two phase shifters PS1 and PS2 are placed after the first plate and before the last plate of the interferometer, respectively, to prepare the initial phases $\vec{\phi}$ and induce the relative phase shifts $\vec{\chi}$. The prepared and induced phases in path 3 and path 1 are controlled by rotations of either PS1 or PS2, respectively, while a combined and equal rotation of both phase shifters controls the phases in path 2. Two indium absorbers of thickness 0.25 mm and 0.5 mm are placed before the second plate and after the third plate, respectively, in order to compensate an inequality in the path intensities. With the foils in place, the relative path intensities were approximately 0.33, 0.32 and 0.34 for path 1, 2 and 3, respectively. Outside of the interferometer, the intensity is measured at the exit ports “+”, “-” and “Aux”. Because of an inevitable offset in the counts at the “-” and “Aux” exit ports, these two intensities are used for preparation procedures and monitoring. The weak values are extracted from measurements at the “+” beam.



3 State preparation

In this work, the state preparation is characterized by the selection of initial phases $\vec{\phi}$ implemented through rotations of PS1 and PS2. For example, let us consider the state

$$|\psi(0, 0, 0)\rangle = \frac{|1\rangle + |2\rangle + |3\rangle}{\sqrt{3}} \quad (8)$$

as the target of our preparation procedure. The interferometer starts in a state of unknown phases $\vec{\phi} = (\phi_1, \phi_2, \phi_3)$ and, up to a global phase, our preparation can be achieved by setting $\phi_1 = \phi_2 = \phi_3$. A measurement performed at the “Aux” beam for different positions of PS 1 can be used to set $\phi_2 = \phi_3$, corresponding to a minimum in the measured intensity. The interferometer state will now be

$$|\psi(\phi_1, \phi_2, \phi_2)\rangle = \frac{e^{i\phi_1} |1\rangle + e^{i\phi_2} (|2\rangle + |3\rangle)}{\sqrt{3}}. \quad (9)$$

Similarly, intensity measurements at the “+” beam for different positions of PS 2 can be used to set $\phi_1 = \phi_3$, corresponding to a maximum in the intensity. This results in a state of equal phases $\phi_1 = \phi_2 = \phi_3$

$$|\psi(\phi_1, \phi_1, \phi_1)\rangle = e^{i\phi_1} \frac{|1\rangle + |2\rangle + |3\rangle}{\sqrt{3}} = |\psi(0, 0, 0)\rangle, \quad (10)$$

which completes the preparation of our target state. In this experiment, a total of 6 preparations are implemented

$$|\psi(0, 0, 0)\rangle = \frac{|1\rangle + |2\rangle + |3\rangle}{\sqrt{3}}, \quad (11)$$

$$|\psi(0, \pi, \pi)\rangle = \frac{|1\rangle - |2\rangle - |3\rangle}{\sqrt{3}}, \quad (12)$$

$$|\psi(0, -\frac{\pi}{2}, -\pi)\rangle = \frac{|1\rangle - i|2\rangle - |3\rangle}{\sqrt{3}}, \quad (13)$$

$$|\psi(0, -\pi, -\frac{\pi}{2})\rangle = \frac{|1\rangle - |2\rangle - i|3\rangle}{\sqrt{3}}, \quad (14)$$

$$|\psi(-\frac{\pi}{3}, 0, \frac{2\pi}{3})\rangle = \frac{e^{-i\frac{\pi}{3}}|1\rangle + |2\rangle + e^{i\frac{2\pi}{3}}|3\rangle}{\sqrt{3}} = \frac{1}{\sqrt{3}} \left(\frac{1-i\sqrt{3}}{2}|1\rangle + |2\rangle - \frac{1-i\sqrt{3}}{2}|3\rangle \right), \quad (15)$$

$$|\psi(0, -\frac{\pi}{3}, \frac{2\pi}{3})\rangle = \frac{|1\rangle + e^{i\frac{2\pi}{3}}|2\rangle + e^{-i\frac{\pi}{3}}|3\rangle}{\sqrt{3}} = \frac{1}{\sqrt{3}} \left(|1\rangle + \frac{1-i\sqrt{3}}{2}|2\rangle - \frac{1-i\sqrt{3}}{2}|3\rangle \right). \quad (16)$$

4 Measurements

Let's consider a general prepared state $|\psi(\vec{\phi})\rangle$ and no induced phase shifts $\vec{\chi} = \vec{0}$. By using the phase shifters, we can increase the initial phase ϕ_j of one path in steps of $\pi/8$, generating the distinct initial states $|\psi(\vec{\phi}_{n,j})\rangle$, with $\vec{\phi}_{n,j} = \vec{\phi} + \frac{n\pi}{8}\vec{\delta}_j$ and $n = \{0, 1, 2, \dots\}$. Measuring the outgoing intensity for different initial states $|\psi(\vec{\phi}_{n,j})\rangle$ at $\vec{\chi} = \vec{0}$ is equivalent to measuring the intensity for the initial state $|\psi(\vec{\phi})\rangle$ with induced phase shifts $\chi_j = \frac{n\pi}{8}$, since

$$\begin{aligned} I(\vec{\phi}_{n,j}) &= A \left| \langle \psi_+ | \psi(\vec{\phi}_{n,j}) \rangle \right|^2 = A \left| \langle \psi_+ | e^{-i\frac{n\pi}{8}\hat{\Pi}_j} | \psi(\vec{\phi}) \rangle \right|^2 = I(\vec{\phi} + \frac{n\pi}{8}\vec{\delta}_j) \\ &= A \left| \langle \psi_+ | \psi(\vec{\phi}) \rangle \right|^2 \left[1 + 2(|w_j|^2 - w_j^R)(1 - \cos \frac{n\pi}{8}) + 2w_j^I \sin \frac{n\pi}{8} \right]. \end{aligned} \quad (17)$$

Therefore, by performing such measurement, for each initial state one can obtain the intensities at $\chi_j = -\frac{\pi}{2}, \frac{\pi}{2}, \pi$ as

$$\begin{aligned} I(\vec{\phi}_{n,j} + \frac{\pi}{2}\vec{\delta}_j) &= I(\vec{\phi}_{n+4,j}) \\ I(\vec{\phi}_{n,j} + \pi\vec{\delta}_j) &= I(\vec{\phi}_{n+8,j}) \\ I(\vec{\phi}_{n,j} - \frac{\pi}{2}\vec{\delta}_j) &= I(\vec{\phi}_{n-4,j}) \end{aligned} \quad (18)$$

An example of this procedure for the initial state $|\psi(0, -\frac{\pi}{2}, -\pi)\rangle$ is shown in Fig. 2; the top, center and bottom plots correspond to the measurement of $I(\vec{\phi}_{n,j}) = I(\vec{\phi} + \frac{n\pi}{8}\vec{\delta}_j)$ for $j = 1, 2$ and 3 , respectively. For instance, in the top plot, the intensity measured at the points $n = 0, 16, 24, 48$ correspond to the initial state $|\psi(0, -\frac{\pi}{2}, -\pi)\rangle$ and $\vec{\chi} = 0$, while the points at $n = 4, 20, 28$ corresponds to the intensity measured for same initial state and $\vec{\chi} = \frac{\pi}{2}\vec{\delta}_1$. Equivalently, the latter point also corresponds to the initial state $|\psi(\frac{\pi}{2}, -\frac{\pi}{2}, -\pi)\rangle$ and $\vec{\chi} = 0$. Note that, the points $n = 0, 16, 24, 48$ correspond to the same initial state $|\psi(\vec{\phi})\rangle$ in all three plots, however, this is not true for the rest of the points.

5 Data correction

The measured intensity, including the loss of contrast due to inevitable imperfections of a real-life set up, is modeled as

$$I^{\text{meas.}}(\vec{\phi}_{n,j}) = \frac{A}{3} \left[1 + \frac{2}{3} C_{12} \cos(\phi_{n,1} - \phi_{n,2}) + \frac{2}{3} C_{13} \cos(\phi_{n,1} - \phi_{n,3}) + \frac{2}{3} C_{23} \cos(\phi_{n,2} - \phi_{n,3}) \right], \quad (19)$$



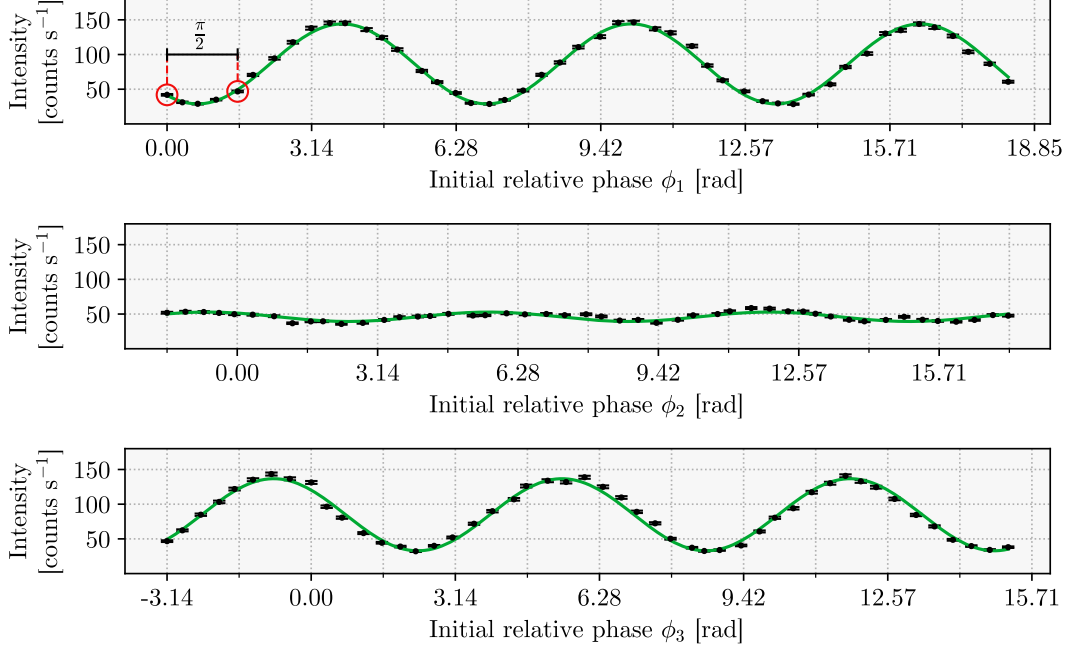


Figure 2: Measurement example for

where C_{jk} , with $j, k = \{1, 2, 3\}$ and $j \neq k$, represent the contrast reduction factor in each two-paths loop made of the paths j and k . Therefore, the corrected intensity can be obtained as

$$I^{\text{corr.}}(\vec{\phi}_{n,j}) = I^{\text{meas.}}(\vec{\phi}_{n,j}) + \frac{A}{3} \left[1 + \frac{2}{3} (1 - C_{12}) \cos(\phi_{n,1} - \phi_{n,2}) + \frac{2}{3} (1 - C_{13}) \cos(\phi_{n,1} - \phi_{n,3}) + \frac{2}{3} (1 - C_{23}) \cos(\phi_{n,2} - \phi_{n,3}) \right]. \quad (20)$$

In order to implement the correction, the contrasts C_{jk} and the relative-path intensities c_j^2 needs to be extracted. This can be achieved from the measurements performed with the initial state $|\psi(0, 0, 0)\rangle$ or $|\psi(0, \pi, \pi)\rangle$. In fact, if we consider a measurement varying ϕ_j starting from the prepared state $|\psi(0, 0, 0)\rangle$, we obtain

$$I_{\phi_1, 0, 0}^{\text{meas.}}(0) = \frac{A}{3} \left(1 + \frac{2}{3} C_{23} + \frac{2}{3} (C_{12} + C_{13}) \cos \phi_1 \right) \quad (21)$$

$$I_{0, \phi_2, 0}^{\text{meas.}}(0) = \frac{A}{3} \left(1 + \frac{2}{3} C_{13} + \frac{2}{3} (C_{12} + C_{23}) \cos \phi_2 \right) \quad (22)$$

$$I_{0, 0, \phi_3}^{\text{meas.}}(0) = \frac{A}{3} \left(1 + \frac{2}{3} C_{12} + \frac{2}{3} (C_{13} + C_{23}) \cos \phi_3 \right) \quad (23)$$

and, since A and c_j are already extracted from the measurement of the single-relative path intensities, the contrast reduction factor C_{jk} can be obtained from the offset of the oscillations. Similarly, it can be extracted, starting from the prepared state $|\psi(0, \pi, \pi)\rangle$ as

$$I_{\phi_1, \pi, \pi}^{\text{meas.}}(0) = \frac{A}{3} \left(1 + \frac{2}{3} C_{23} - \frac{2}{3} (C_{12} + C_{13}) \cos \phi_1 \right) \quad (24)$$

$$I_{0, \phi_2, \pi}^{\text{meas.}}(0) = \frac{A}{3} \left(1 - \frac{2}{3} C_{13} + \frac{2}{3} (C_{23} - C_{12}) \cos \phi_2 \right) \quad (25)$$

$$I_{0, \pi, \phi_3}^{\text{meas.}}(0) = \frac{A}{3} \left(1 - \frac{2}{3} C_{12} + \frac{2}{3} (C_{23} - C_{13}) \cos \phi_3 \right). \quad (26)$$

Since the contrast reduction depends on the environment conditions, the correction for a general measurement $I^{\text{meas.}}(\vec{\phi}_{n,j})$ is implemented using the contrast reduction factors C_{jk} extracted from the closest measurements with preparation $|\psi(0, 0, 0)\rangle$ or $|\psi(0, \pi, \pi)\rangle$.

6 Results



The best results obtained for each preparation are presented. Each figure includes the results obtained without data correction (top) and with the data correction (bottom).

$$|\psi(0, 0, 0)\rangle = \frac{|1\rangle + |2\rangle + |3\rangle}{\sqrt{3}}$$

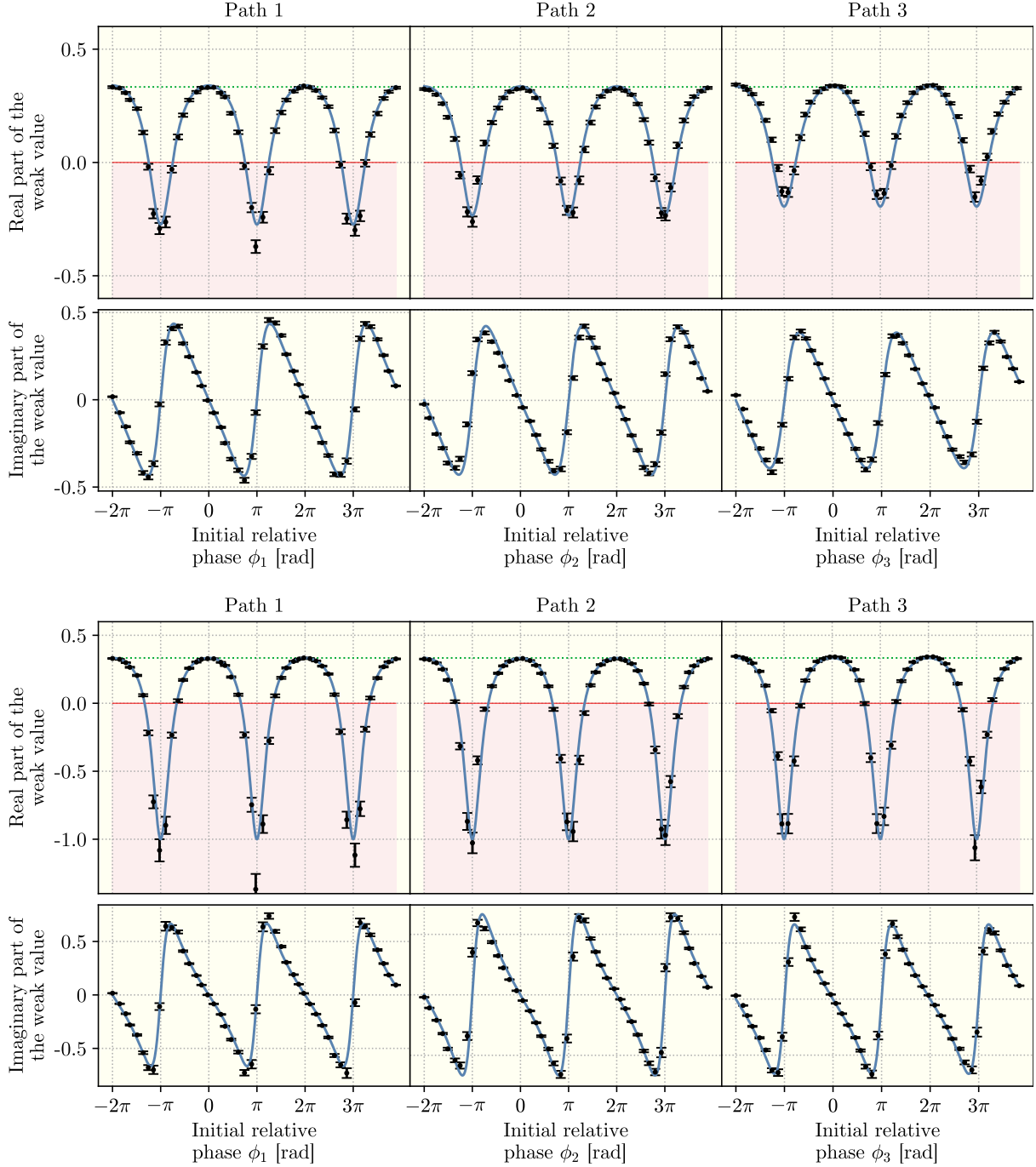


Figure 3: Measurement results uncorrected (top) and corrected (bottom)

$$|\psi(0, \pi, \pi)\rangle = \frac{|1\rangle - |2\rangle - |3\rangle}{\sqrt{3}}$$

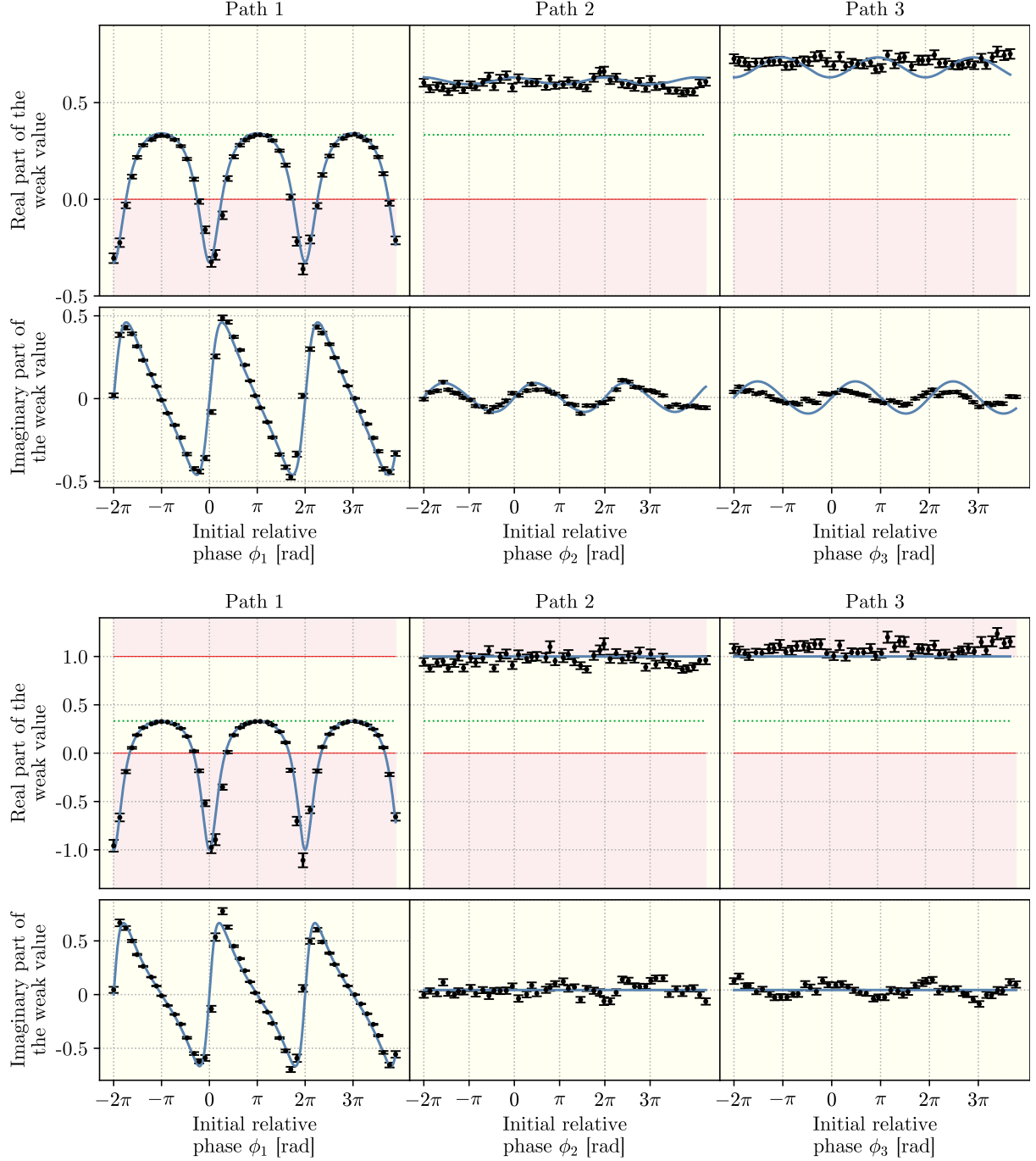


Figure 4: Measurement results uncorrected (top) and corrected (bottom)

$$|\psi(0, -\frac{\pi}{2}, -\pi)\rangle = \frac{|1\rangle - i|2\rangle - |3\rangle}{\sqrt{3}}$$

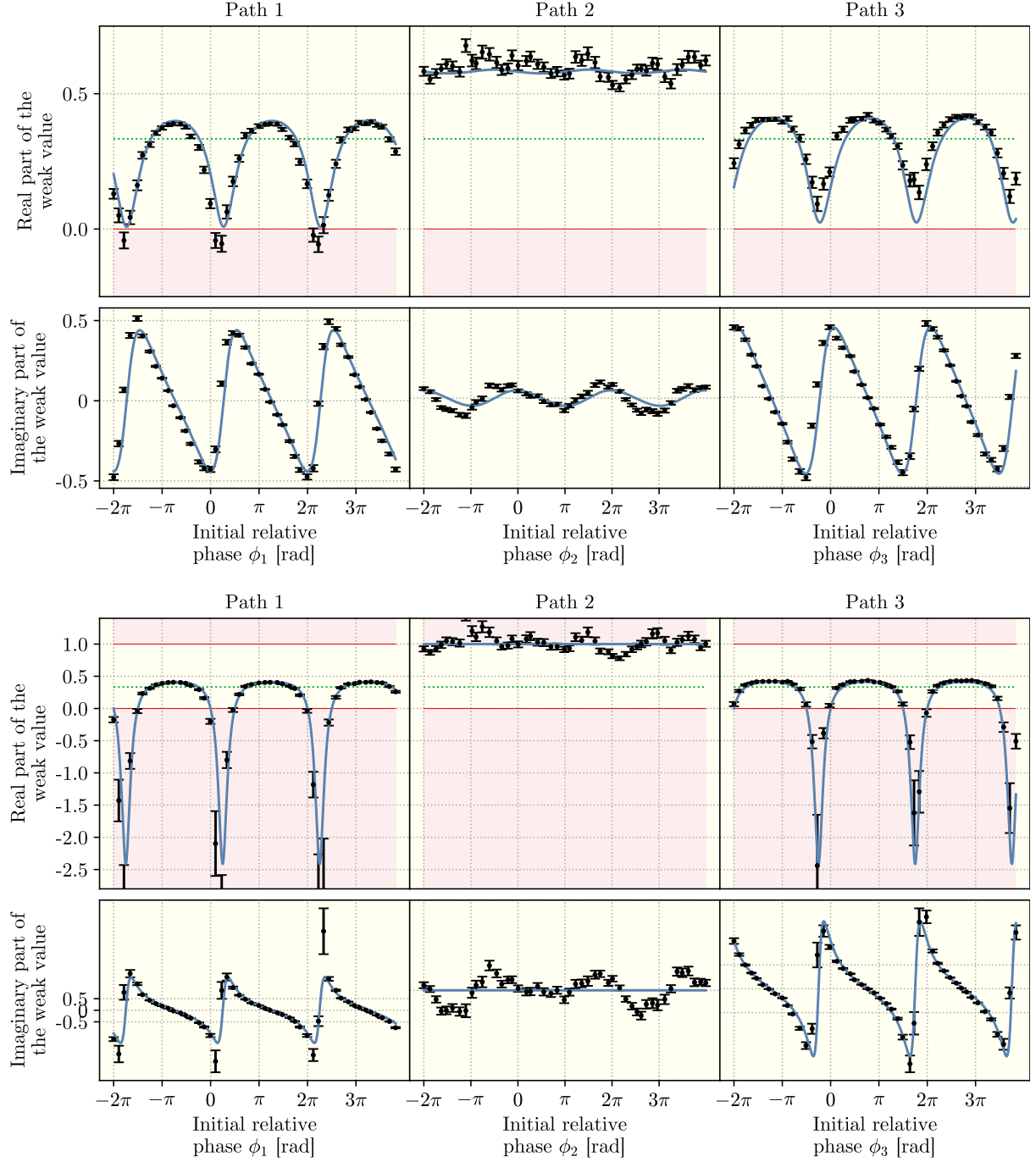


Figure 5: Measurement results uncorrected (top) and corrected (bottom)

$$|\psi(0, -\pi, -\frac{\pi}{2})\rangle = \frac{|1\rangle - |2\rangle - i|3\rangle}{\sqrt{3}}$$

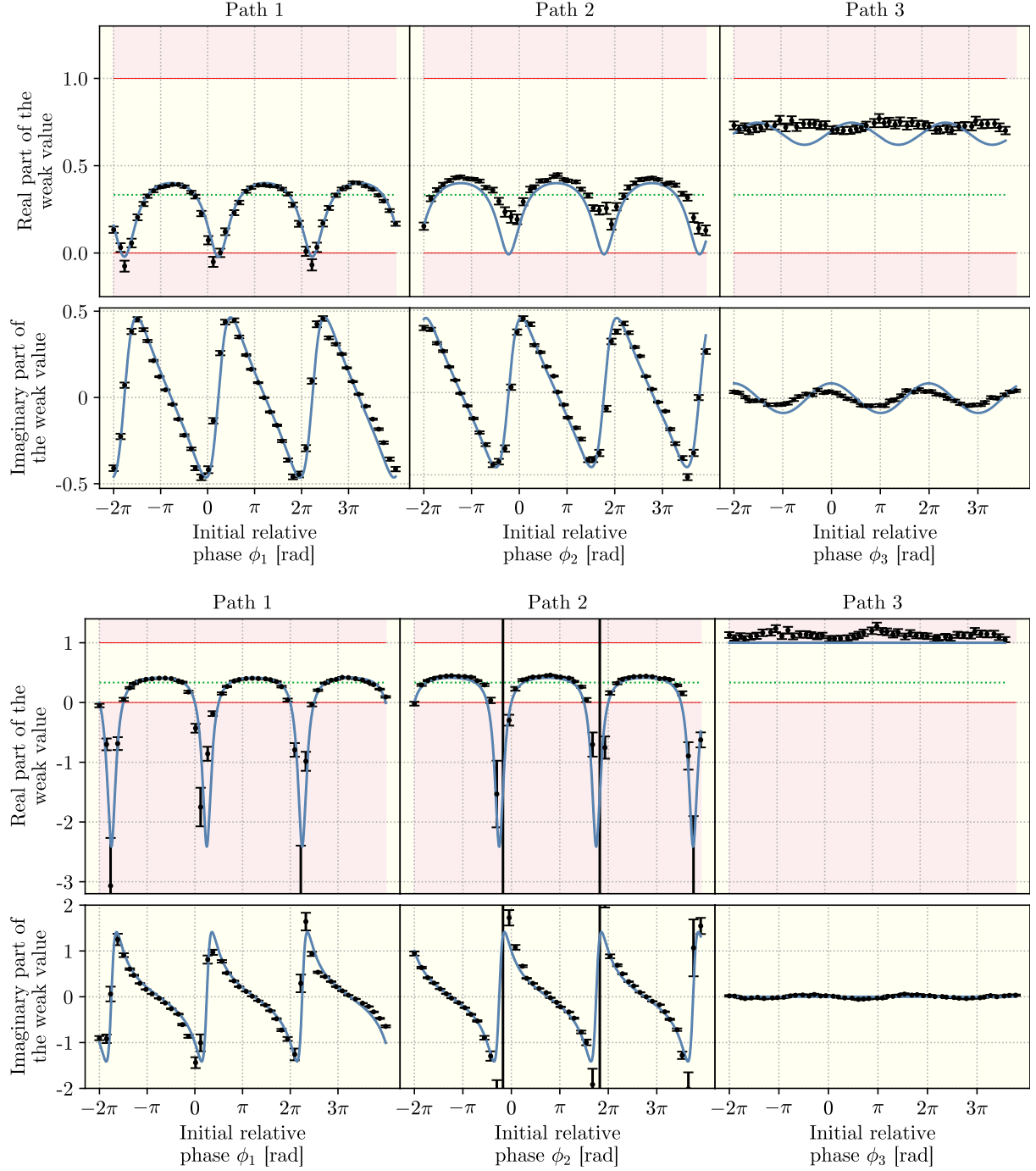


Figure 6: Measurement results uncorrected (top) and corrected (bottom)

$$|\psi(-\frac{\pi}{3}, 0, \frac{2\pi}{3})\rangle = \frac{e^{-i\frac{\pi}{3}} |1\rangle + |2\rangle + e^{i\frac{2\pi}{3}} |3\rangle}{\sqrt{3}}$$

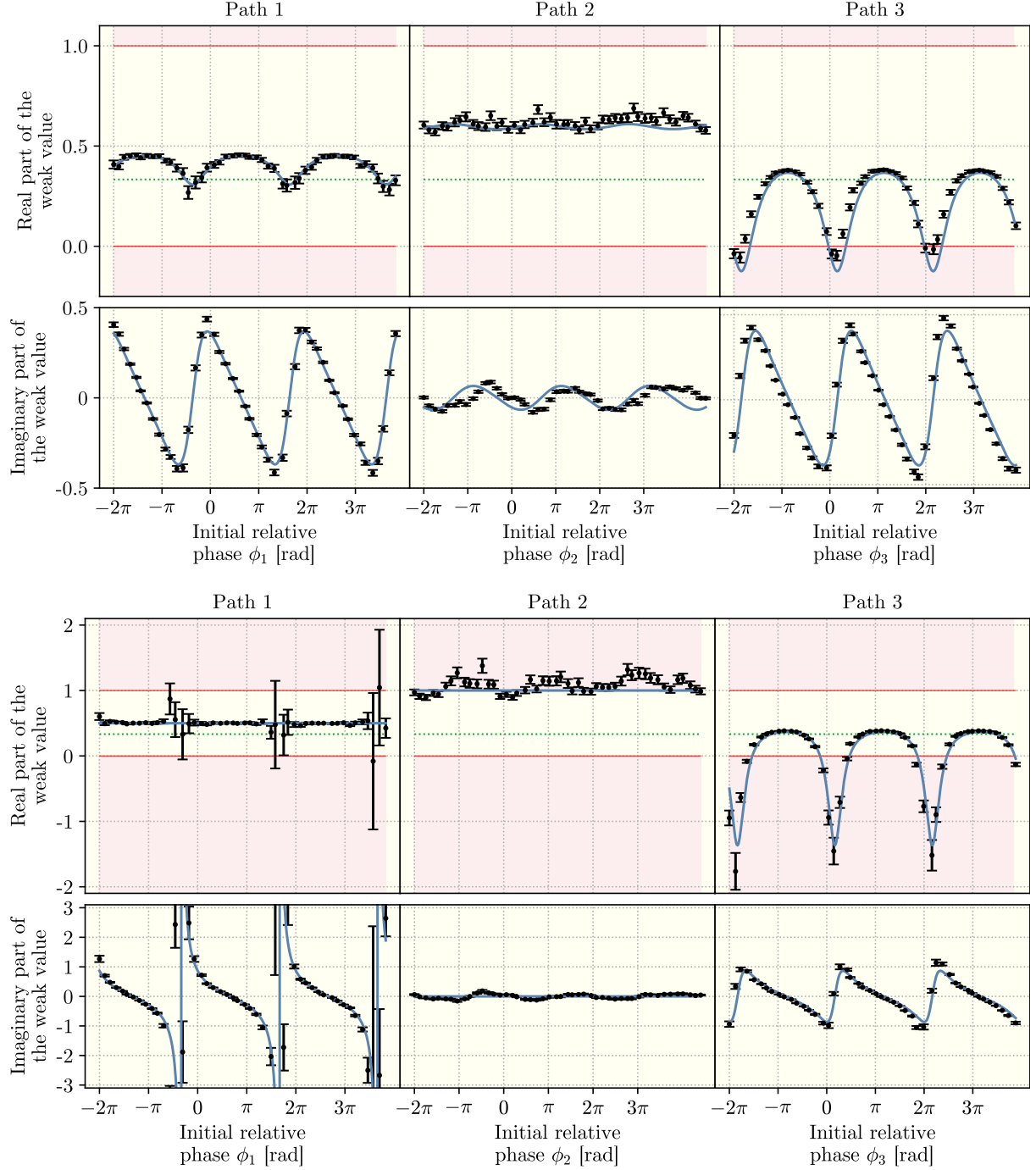


Figure 7: Measurement results uncorrected (top) and corrected (bottom)

$$|\psi(0, -\frac{\pi}{3}, \frac{2\pi}{3})\rangle = \frac{|1\rangle + e^{i\frac{2\pi}{3}}|2\rangle + e^{-i\frac{\pi}{3}}|3\rangle}{\sqrt{3}}$$

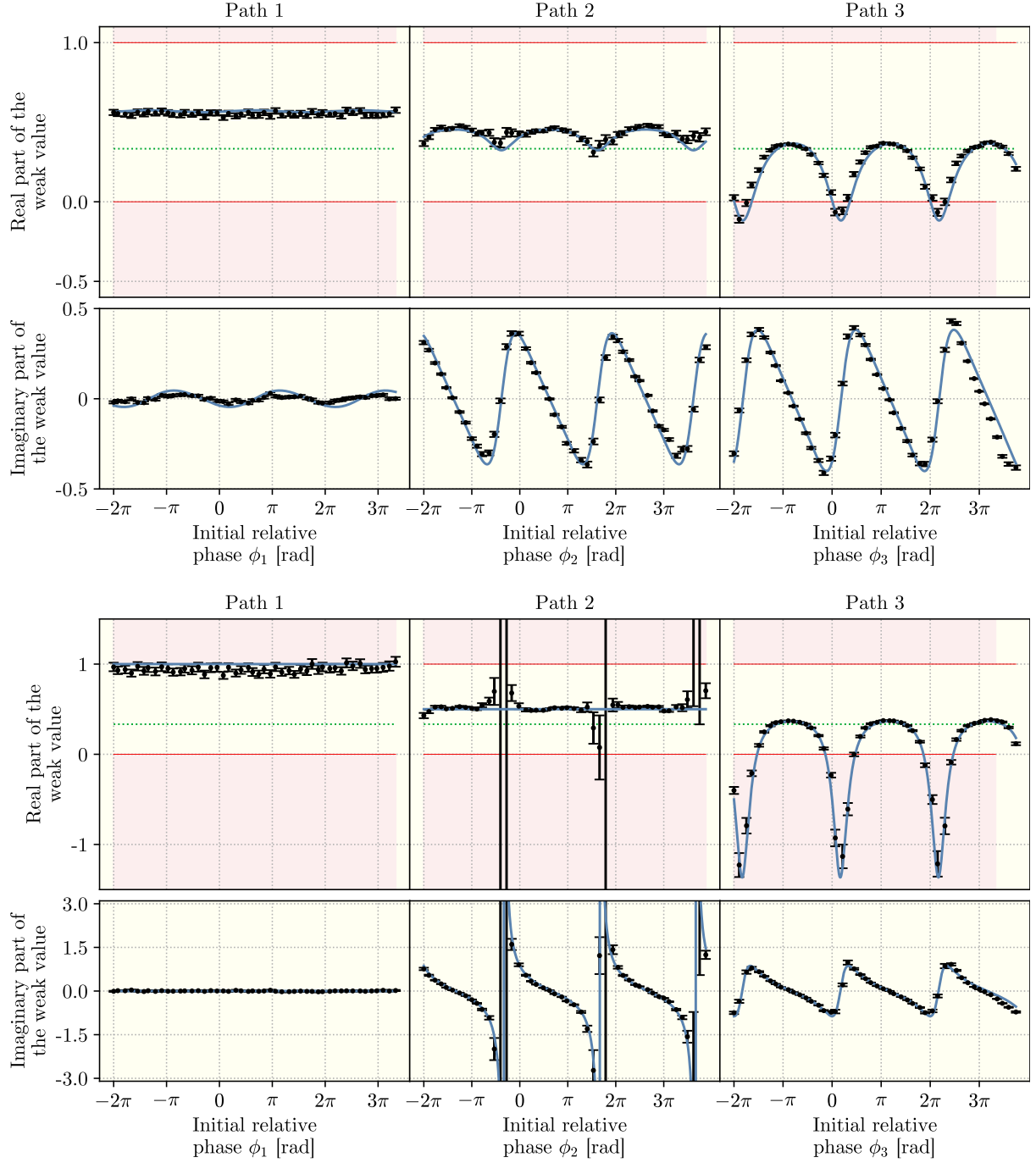


Figure 8: Measurement results uncorrected (top) and corrected (bottom)

References

- [1] Masiello, I. V., Lemmel, H., Dvorak, A., Sponar, S. & Hasegawa, Y. Anomalous weak values in a generalized mach-zehnder interferometer extracted directly from intensity measurements. *arXiv preprint arXiv:2606.24798* (2026).